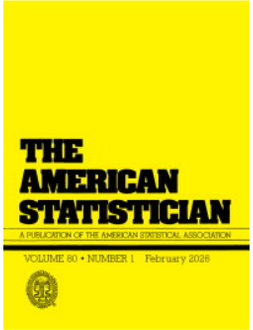

The American Statistician

ISSN: 0003-1305 (Print) 1537-2731 (Online) Journal homepage: www.tandfonline.com/journals/utas20

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To cite this article: Toby Hackmann, Doranne Thomassen, Anne M. Stiggelbout, Saskia le Cessie, Hein Putter, Liesbeth C. de Wreede, Ewout W. Steyerberg & on behalf of the 4D PICTURE Consortium (2026) Effective Sample Size for the Kaplan-Meier Estimator: A Valuable Measure of Uncertainty?, The American Statistician, 80:1, 100-108, DOI: 10.1080/00031305.2025.2542390

To link to this article: https://doi.org/10.1080/00031305.2025.2542390

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THE AMERICAN STATISTICIAN
2026, VOL. 80, NO. 1, 100–108: Statistical Practice
https://doi.org/10.1080/00031305.2025.2542390

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Effective Sample Size for the Kaplan-Meier Estimator: A Valuable Measure of Uncertainty?

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ABSTRACT

Sample size is an essential indicator of the uncertainty in clinical research results. When studies present time-to-event outcomes with Kaplan-Meier curves, these are often accompanied by the remaining number of patients at risk in a table below the curve. The number at risk at time t informs about uncertainty of the hazard at t, rather than the uncertainty of the estimated survival probability until t, $\hat{S}(t)$. We aim to review the role of the effective sample size of $\hat{S}(t)$ to reflect the uncertainty in survival probability estimation. Effective sample size is defined as the size of a hypothetical sample with complete follow-up until time t, that would give the same variance as the variance of the Kaplan-Meier estimate $\hat{S}(t)$. We consider hypothetical scenarios and a publicly available dataset with patients treated for colon cancer. These illustrations support that effective sample size provides a readily interpretable measure of uncertainty for survival curves in the presence of censoring. We show that effective sample size can also quantify the loss of information when the reporting for an ongoing study is moved to an earlier time point. In conclusion, effective sample size is a valuable measure of uncertainty in survival analysis.

ARTICLE HISTORY
Received March 2025
Accepted July 2025

KEYWORDS
Kaplan-Meier; Risk communication; Sample size; Survival; Uncertainty

1. Introduction

Uncertainty in results from clinical studies is lower with larger sample sizes. Sample size hence is an intuitive measure of uncertainty (Sedlmeier and Gigerenzer 1997). For survival, the time of the event of interest is typically not observed for all study participants ("censoring"). Censoring may occur for two main reasons: patients can drop out of the study ("lost to follow-up"), or they can be administratively censored when the study observation period ends. In both cases, a common assumption in survival analysis is that censoring at time t provides no additional information about prognosis after t (Fojo and Simon 2021). In medical research and related fields, the Kaplan-Meier product limit estimator is widely used to provide survival probabilities over time while respecting censoring (Kaplan and Meier 1958). Kaplan-Meier curves are often accompanied by the remaining number of patients at risk in a table below the curve. The number at risk at time t informs about uncertainty of the hazard at t, and survival after t.

In this article we focus on the uncertainty of the estimated survival probability until t. We aim to examine the value of "effective sample size" for this uncertainty. Effective sample size is defined as the sample size equivalent to the sample size of a study without censoring (Kaplan and Meier 1958). Although, the concept of effective sample size has been discussed with varying definitions and purposes (Cutler 1947; Cutler and Ederer 1958; Peto et al. 1977; Rothman 1978; Dorey and Korn 1987; Meier et al. 2004), it has not often been used as a measure to communicate the uncertainty of estimated survival probabilities. We illustrate effective sample size in a case study and in simulations to better understand its behavior and potential roles in survival analysis. We end with a discussion on utility of the effective sample size and future research.

2. Methods

2.1. Effective Sample Size for the Kaplan-Meier Estimate of Survival

Effective sample size has been defined by Kaplan and Meier (1958) as a sample size, "which in the absence of losses would give the same variance [as the variance of the Kaplan-Meier

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Supplementary materials for this article are available online. Please go to www.tandfonline.com/r/TAS.
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estimate]". Suppose we have estimated the survival probability at the time of interest $\hat{S}(t)$ using the Kaplan-Meier estimator $\hat{S}(t) = \prod_{i:t_i \le t}(1 - d\hat{\Lambda}(t_i))$, where $\hat{\Lambda}(t_i)$ is the Nelson-Aalen estimator of the cumulative hazard. The variance of $\hat{S}(t)$ can be consistently estimated by Greenwood's formula (Greenwood 1926; Andersen et al. 1993). The effective sample size at time t is defined as the size of a hypothetical sample with complete follow-up (no censoring) until time t, with mean survival $\bar{S}(t)$ equal to $\hat{S}(t)$, such that the variance of $\hat{S}(t)$ is the same as the sampling variance of $\bar{S}(t)$ in the hypothetical population. To calculate this effective sample size, we equate the estimated variance of the Kaplan-Meier estimated survival outcome $\hat{S}(t)$ with the binomial sampling variance of the mean outcome $\bar{S}(t)$.

$$\hat{V}_{GW}\left[\hat{S}(t)\right] = V_{\text{Bin}}\left[\bar{S}(t)\right], \tag{1}$$

which yields

$$\hat{S}^2(t)\sum_{t_i \le t}\frac{\delta_i}{R_i\left(R_i - \delta_i\right)} = \frac{\bar{S}(t)\left(1 - \bar{S}(t)\right)}{N_{\text{eff}}}, \tag{2}$$

with $t_1 < \cdots < t_D$ the D distinct times where events are observed and δ_i are the number of events and R_i the number of patients at risk at time t_i. The Greenwood estimator for the variance of the Kaplan-Meier estimator is used on the left hand side (Greenwood 1926). Now we replace the sample mean survival $\bar{S}(t)$ with the estimated survival $\hat{S}(t)$. Now (2) can be rewritten to provide the estimator for effective sample size:

$$\hat{N}_{\text{eff}}(t) = \frac{1 - \hat{S}(t)}{\hat{S}(t)\sum_{i:t_i \le t}\frac{\delta_i}{R_i(R_i - \delta_i)}}. \tag{3}$$

The estimator for effective sample size is undefined before the first event since $\sum_{i:t_i \le t}\frac{\delta_i}{R_i(R_i-\delta_i)} = 0$. Before the first event, we define effective sample size as the number of patients at risk of an event.

This quantity is the "effective sample size for estimates based on the Kaplan-Meier estimator," but for brevity and readability we refer to this as N_{eff} or "effective sample size".

2.2. Modified Effective Sample Size for the Kaplan-Meier Estimate of Survival

The Kaplan-Meier survival estimate with Greenwood variance and resulting confidence intervals only change at times with events. After the last event the standard N_{eff} estimator (3) is constant at time points where no events occur but patients are censored. We consider this a limitation of the above definition of N_{eff}, because, intuitively, censoring should decrease the effective sample size. Methods have been proposed to improve the variance and confidence interval estimation in the tail (Peto et al. 1977). One such method is to modify the variance estimation procedure when there is censoring but no event (Dorey and Korn 1987). This method provides better coverage but is also more conservative than the standard Greenwood estimator (Yuan and Rai 2011).

Using the modified variance suggested by Dorey and Korn (1987), instead of Greenwood's variance, we derive a modified effective sample size $N_{\text{eff,mod}}$ (mathematical derivation in Supplement 2). $N_{\text{eff,mod}}$ decreases at times t with only censoring and therefore decreases after the last observed event. We refer to modified effective sample size with $N_{\text{eff,(mod)}}$.

2.3. Number at Risk is the Effective Sample Size of the Hazard

The commonly reported number at risk can also be interpreted as an effective sample size, but of the hazard instead of the survival probability. Nonparametric estimation of the hazard is typically established by focusing on the cumulative hazard. For instance, the Nelson-Aalen estimator has an increment at each of the event time points t_i of size $d\hat{\Lambda}(t_i) = \frac{\delta_i}{R_i}$, where δ_i and R_i are the number of events and the number at risk, respectively, at time t_i. Using Greenwood's formula, the estimate of the variance of this hazard increment equals $\frac{(R_i-\delta_i)\delta_i}{R_i^3}$ (Andersen et al. 1993). This variance equals the binomial variance $\frac{p_i(1-p_i)}{n}$, with $p_i = \frac{\delta_i}{R_i}$, and $n = R_i$ the effective sample size. So, indeed, the number at risk is the effective sample size of the nonparametric cumulative hazard estimate increment at each time point.

2.4. Illustrative Clinical Data

We illustrated the behavior of effective sample size using a publicly available dataset with the survival package in R (Therneau 2024). The dataset includes patients from a clinical trial of adjuvant chemotherapy after a resection with curative intent for a histologically confirmed adenocarcinoma of the colon or rectum (Laurie et al. 1989; Moertel et al. 1995). Patients were randomized to either observation (Obs), treatment with levimasole (Lev) alone, or treatment with levimasole and fluorouracil (Lev + 5FU). Overall survival and recurrence-free survival were end points. We focused our analyses on overall survival (Figure 1(A)).

2.5. Simulation Study: Dependence on Hazard Functions

We examined the behavior of $N_{\text{eff,(mod)}}$ further in some simulated settings. In the first simulation we varied hazards for events and censoring in the simsurv package (Brilleman et al. 2021). We generated survival times T_E and censoring times T_C from Weibull hazard functions. An event was observed at time T_E if $T_E \le T_C$ and a patient was censored at time T_C if $T_C < T_E$. All patients were administratively censored at $t = 5$. We generated survival data using Weibull(a, b) distributions, with a representing the shape parameter, and b the scale parameter, in such a way that the cumulative hazard was

$$\Lambda(t) = \left(\frac{t}{b}\right)^a. \tag{4}$$

Three different Weibull distributions were used for the event and censoring time generation, yielding a total of nine combinations. Shape parameters (a) were chosen such that events/censoring occurred early $(a = 4)$, evenly spread $(a = 1)$, or late $(a = \frac{1}{4})$ during follow-up. The scale parameters (b), defining when the cumulative hazard reaches 1, was set to $b = 2.5$ for event simulations, while $b = 3.5$ was used for censoring (Figure S1 in Supplement 3). This difference ensures that, on average, more patients will have an event than be censored with the same shape parameter. Each simulated sample consisted of 1000 patients. The estimated survival function, $N_{\text{eff,(mod)}}$, the number of patients uncensored and the number

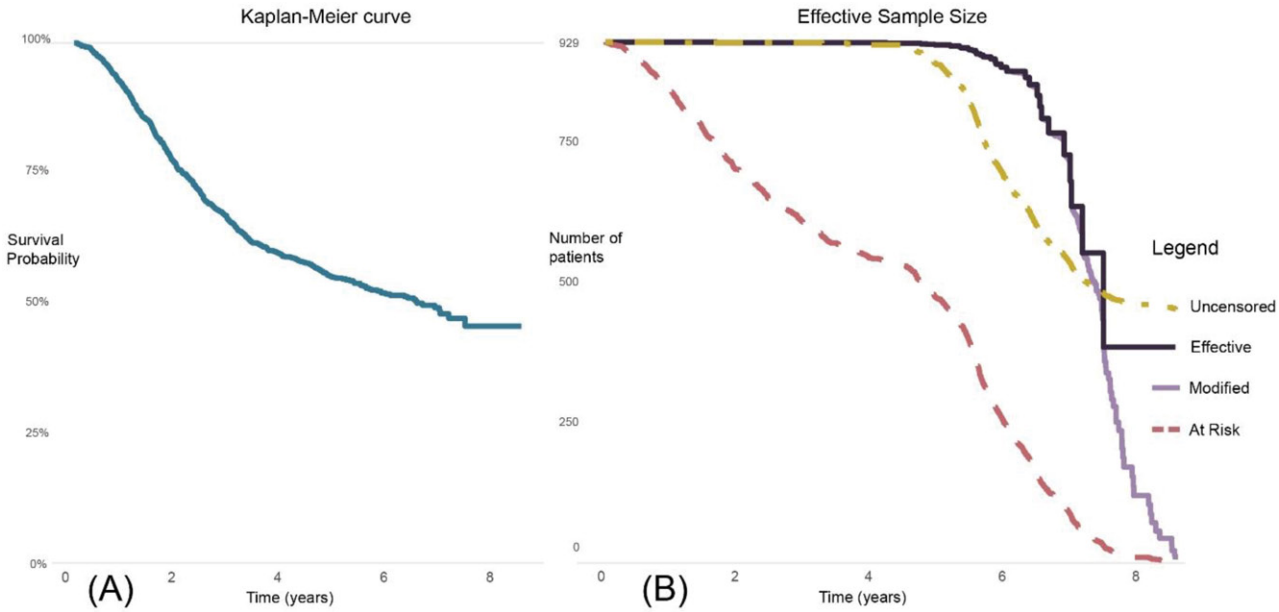

Figure 1. Kaplan-Meier Estimate of overall survival based on the Colon cancer dataset (a), with corresponding indicators of sample size (b). Effective sample size (standard or modified) is always higher than the number at risk an only begins to decrease after patients are censored. No censoring occurred before 5 years of follow-up ($t = 5$).

at risk were averaged over 50,000 replications for stability. The number at risk is defined as the number of patients who were both alive and still in follow-up. The number not censored is defined as the sum of the number at risk and the number of previous events.

2.6. Quantifying Loss of Information for Early Study Reporting

We investigated the potential role of effective sample size in informing about the loss of information in reporting the results of a cohort study or trial early, with illustration in the colon data. The outcome of interest was the 5-year overall survival in the three arms of the trial. For this illustration, we generated enrollment times and event times on the scale of calendar time of the study (Supplement 4).

All analyses were conducted in R (R Core Team 2003) version 4.3.1 and specifically the survival package (Therneau 2024). The functions used to estimate the effective sample size work on the survfit object of (stratified) Kaplan-Meier models and can be found on Github (*https://github.com/toby-hackmann/ess-kaplan-meier*).

3. Results

3.1. Illustration in Colon Cancer Data

Overall survival was 46% [95% confidence interval 41%–51%] at 8 years in the colon cancer dataset with 452 events observed among 929 patients (Figure 1(A)). Before any patient was censored, $N_{eff,(mod)}$ was 929, equal to the total sample size. After patients were censored, $N_{eff,(mod)}$ began to decrease. The number at risk was much smaller than $N_{eff,(mod)}$ at later times t (Figure 1(B)). The number not censored decreased earlier than $N_{eff,(mod)}$, because of events occurring earlier than censoring.

The Kaplan-Meier estimated survival curve was constant after ~8 years (Figure 1(A)), yielding a constant estimate of N_{eff} in the tail while—by definition—$N_{eff,mod}$ decreased with continued censoring of patients. At 8 years, N_{eff} for the overall survival estimate was 385 and $N_{eff,mod}$ was 349, while the number at risk was only 26.

We note that N_{eff} at the tail of the curve was highly dependent on the timing of the last event in the three treatment groups ($t = 8$ Figures 1(B) and 2). The cohort with Lev + 5FU has the earliest last event and the highest N_{eff} (210) at $t = 8$, and the cohort with only Lev has the latest last event and the lowest N_{eff} (75). $N_{eff,mod}$ avoided this dependency, with more similar effective sample sizes at $t = 8$ (75–103) than N_{eff} (75–210, Figure 2).

3.2. Dependence of Modified N_{eff} on Hazards of Events and Censoring

In the simulation scenario of early censoring (left column Figure 3) $N_{eff,(mod)}$ decreased more rapidly. As time continued, $N_{eff,(mod)}$ remained close to the number of uncensored patients. Patients censored early during follow-up had a large impact on $N_{eff,(mod)}$, which decreased by approximately 1 per censored patient, while the impact of censoring on $N_{eff,(mod)}$ in the tail was lower. In the simulation scenario with a constant hazard of censoring an approximately linear decrease in effective sample size over time occurred (middle column Figure 3). The late censoring scenario (right column Figure 3), resembled the colon data with most events occurring early in the follow-up. Initially, $N_{eff,(mod)}$ was close to the complete sample size. At later times t, $N_{eff,(mod)}$ decreased drastically, to numbers far below the number of uncensored patients. Late censoring combined with early events showed the largest difference between N_{eff} and $N_{eff,\,mod}$.

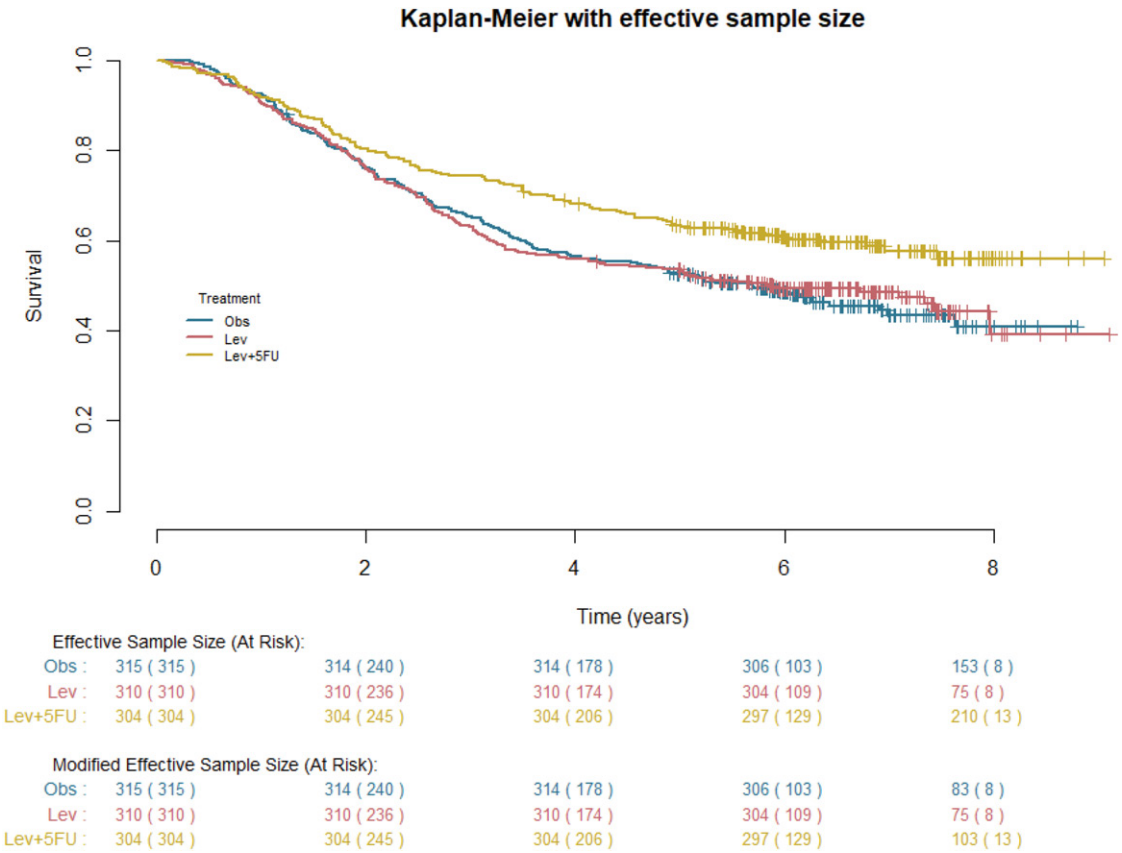

Figure 2. Kaplan-Meier Curve of the Colon data stratified by treatment arm. Effective sample sizes (standard n_{eff} or modified $n_{eff,(mod)}$) provide higher values than the number at risk ("At risk"). While there is no substantial difference in censoring patterns, n_{eff} is quite different for the three curves at $t = 8$ due to the specific times at which the last event occurred. $N_{eff,(mod)}$ is rather similar for the three curves at $t = 8$.

Comparisons between the rows in Figure 3 illustrate how censoring hazards cause different patterns of $N_{eff,(mod)}$ for the survival estimate. There are also differences in the numbers at risk between these scenarios at most times t. However, at a specific time point t_*, the number at risk may be exactly the same while there still are clear differences in $N_{eff,(mod)}$ (Figure S3 in Supplement 5).

3.3. Effective Sample Size to Guide Study Reporting

Effective sample size can inform how much information is available for estimating the survival probability at the time point t of interest. Maximum information is available when all patients have either experienced an event or reached the time point of interest event-free as uncensored observations.

For the colon data, we focus on the 5-year survival estimate $t = 5$. Five years after start of the study, the first enrolled patient may have sufficient follow-up for a $t = 5$ survival estimate. When accrual lasts 4 years, the last enrolled patient has 5 years of follow-up at year 9 after start of the study. We note that 95% of the effective sample size is already achieved at year 8 after start of the study, that is one year before the last patient reaches $t = 5$ (Figure 4). This pattern of available information was confirmed for other trial simulations (Figure S5 in Supplement 6).

4. Discussion

Sample size is a challenging concept for time-to-event data, because not all patients may have complete follow-up (censoring). The concept of effective sample size can provide a measure with the same interpretation as sample size in standard statistical analyses of binary outcomes. We found that the pattern of the effective sample size depended on the hazard for censoring and the hazard for events. How rapidly effective sample size decreases depends especially on the hazard for censoring, with early censoring making survival estimates for later time points less reliable. We consider the modified version of effective sample size $N_{eff,mod}$ attractive for a more natural behavior with Kaplan-Meier estimates with late censoring without late events, because $N_{eff,mod}$ also decreases when there is only censoring. We discuss below why $N_{eff,(mod)}$ might be valuable to add to Kaplan-Meier curves, for risk communication and for study reporting decisions.

4.1. Effective Sample Size below Plots of Kaplan-Meier Curves

Guidelines and tutorials on the presentation of survival curves state that the number of patients at risk should be presented below the Kaplan-Meier curve (Pocock, Clayton, and Altman 2002; Jager et al. 2008; Morris et al. 2019; Vickers et al. 2020).

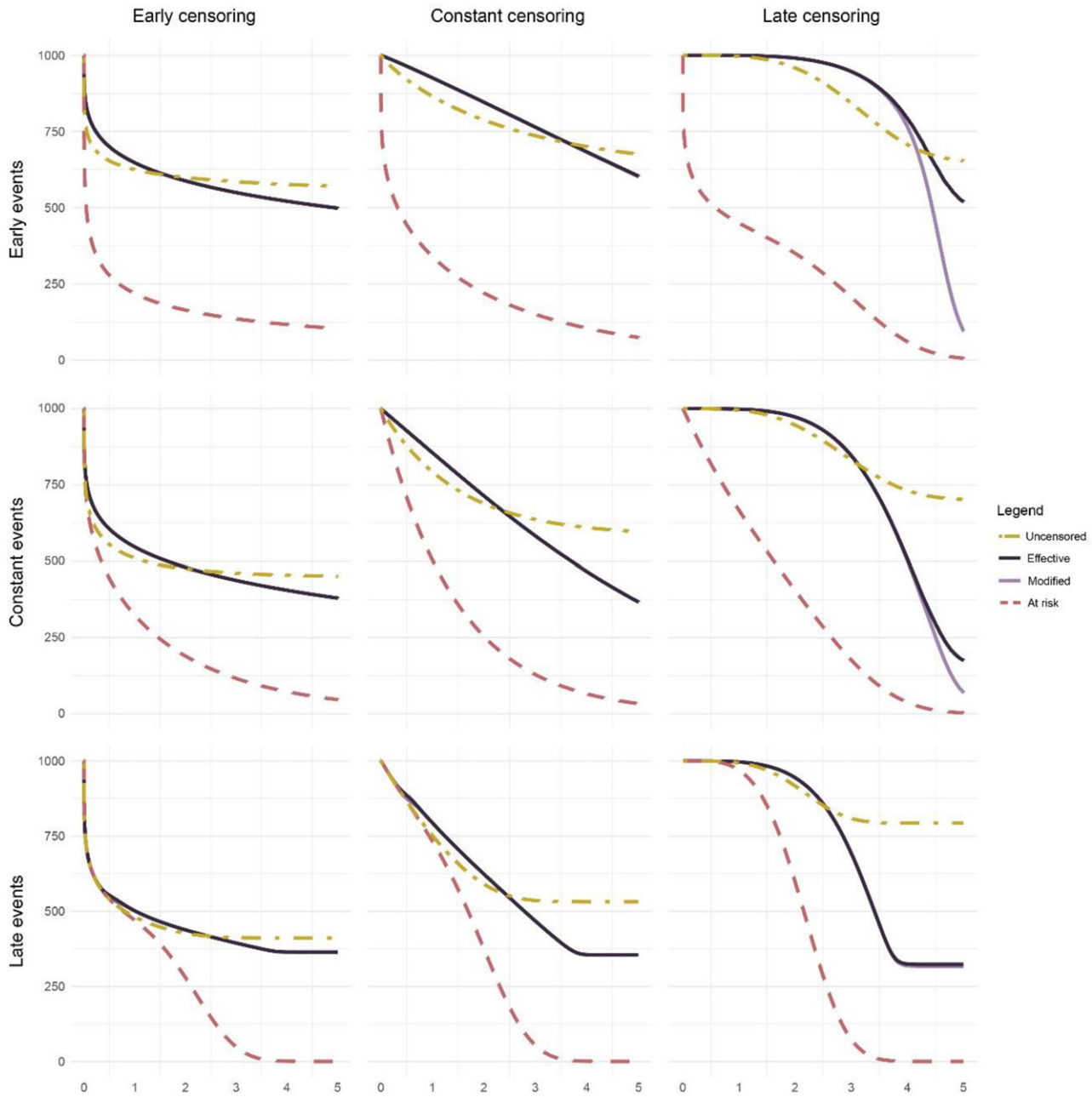

Figure 3. Results of simulations with combinations of for early, constant and late events and censoring patterns. The underlying hazard functions are in supplement Figure 1.

Often this is accompanied by a recommendation that the survival curve is truncated when the number of patients at risk falls below a certain percentage of the number of non-censored patients (e.g., 10% or 20%) (Pocock, Clayton, and Altman 2002), or below an absolute number (e.g., $n < 5$, or $n < 10$) (Vickers et al. 2020). Number at risk is a measure of the reliability of the estimate of the hazard at time t, which relates to the uncertainty for future survival probability estimates beyond time t. In contrast, $N_{eff,(mod)}$ indicates the reliability of the estimated survival probability at time t, which is essential information for patients, physicians and other decision-makers. The contrast in absolute numbers can be large. For example, in the colon data the number at risk at 8 years of follow-up was only 26, while N_{eff} of the estimated survival probability was 385 and $N_{eff,mod}$ was 349.

Effective sample size hence can be useful in addition to or instead of the number of patients at risk in a table below the curve. One might object that effective sample size provides information that is already included in the variance and reflected in a 95% confidence interval. Effective sample size however may be more intuitive for communication of uncertainty with patients and other stakeholders because it is a single number with an explainable interpretation. Additionally, N_{eff} is a strictly decreasing measure with increasing censoring while variance of the Kaplan-Meier estimates is maximized at a survival probability of 50%. This allows for an intuitive interpretation of N_{eff}, where higher is always better. Further empirical studies are needed to assess this claim, extending previous comparative research on presentations of Kaplan-Meier curves (Morris et al. 2019).

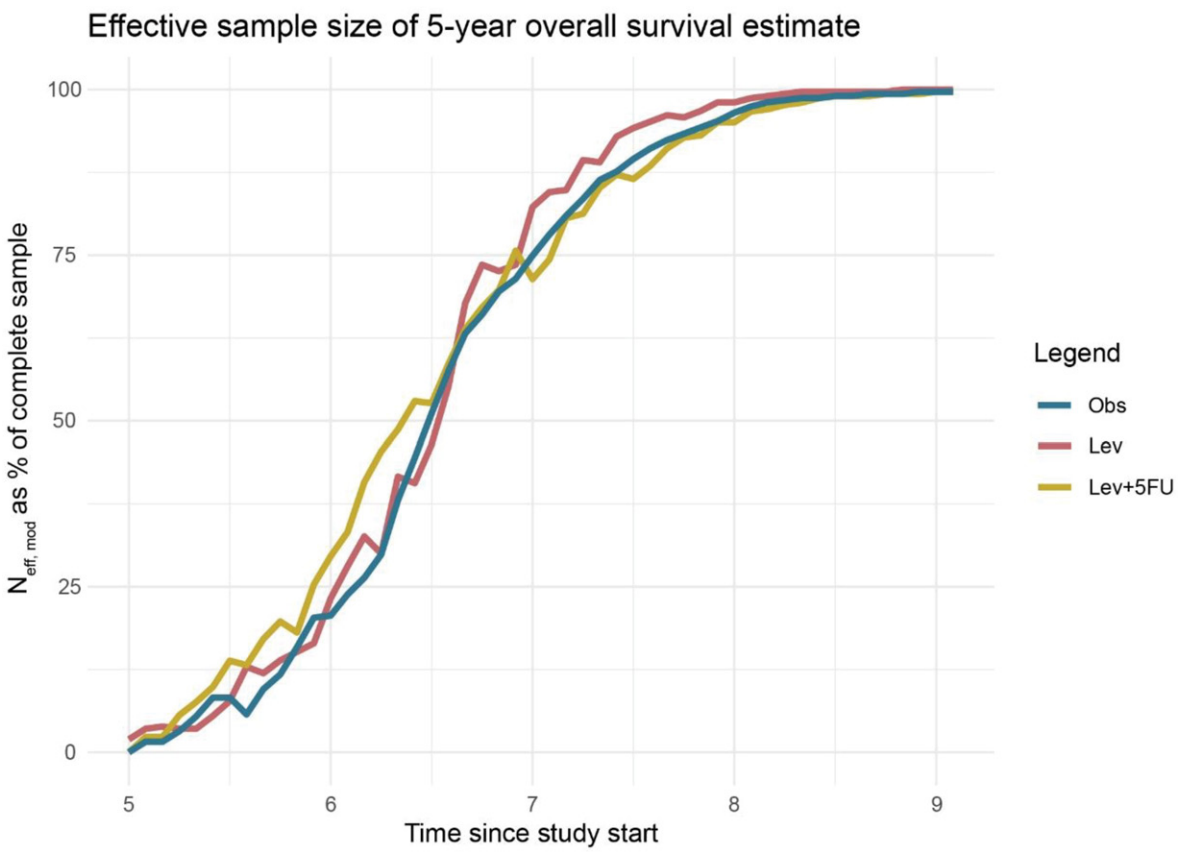

Figure 4. Illustration of modified effective sample size $n_{eff,mod}$ of the 5-year overall survival Kaplan-Meier estimate at increasing trial reporting times in the Colon data stratified by treatment arm. With accrual over 4 years and 5-year follow-up for the last patient, we note that 95% of the information in the study is already available at year 8, one year before the planned end of the study.

4.2. Effective Sample Size for Risk Communication

Open communication about uncertainty is an important step in improving trustworthiness in scientific results (Spiegelhalter 2017a,b). Some claimed that it is futile to try to explain uncertainty in predictions to individual patients (Kattan 2011). Also, domain experts, such as physicians, may be hesitant to communicate uncertainty, because such communication may undermine trust in their work (Fischhoff 2012). This is disputed by others (van der Bles et al. 2020), while indeed people may be less likely to base decisions on numbers with low quality of evidence (Schneider et al. 2022).

In a well-cited study, the instruction with a nomogram for prostate cancer risk states that physicians should tell a patient: "if we had 100 men exactly like you, we would expect between *risk*% − 10% and *risk*% + 10% to remain free of their disease at 5 years" (Kattan and Scardino 2002). Such a presentation may well convey the absolute risk to a patient but the presentation suggests that the prediction is based on 100 patients. This nomogram was based on a Cox proportional hazards model. Here, a consideration is that patients with exceptional covariate patterns have low effective sample, as recently illustrated for predictions from logistic regression models (Thomassen et al. 2024). Moreover, research into effective sample sizes of predictions from machine learning models warns that increased model flexibility reduces effective sample sizes of individual predictions (Thomassen et al. 2025). Future research is needed to define and examine the effective sample size for survival predictions $\hat{S}(t, X_{new})$ based on combinations of covariates, as commonly modeled with Cox regression analysis.

Shared decision making between physicians and patients is increasingly emphasized in modern medicine (Stiggelbout, Pieterse, and De Haes 2015; Rietjens et al. 2024). Communication about uncertainty of the evidence supporting treatment choices is important for the autonomy of patients (Parascandola, Hawkins, and Danis 2002). In an empirical study, reference to uncertainty was noted in only half of consultations with probabilities mentioned (Engelhardt et al. 2017). Only 21% of those references to uncertainty were about the imprecision of the estimate. Effective sample size could help with communication of such imprecision, or epistemic uncertainty (Calin-Jageman and Cumming 2019).

As an example, we show a possible communication tool based on the colon data with effective sample size (Figure 5). For the key trial result, the definition of "patients like you" refers to randomized patients who met the in- and exclusion criteria of the trial. For clinical application, in a real-world setting, important prognostic factors should be considered, and the format for presentation may need further refinement.

We expect that uncertainty communication using effective sample size may not be preferred by all patients. Patients with high numeracy and statistical literacy may appreciate the 95% confidence interval for a survival prediction, while other

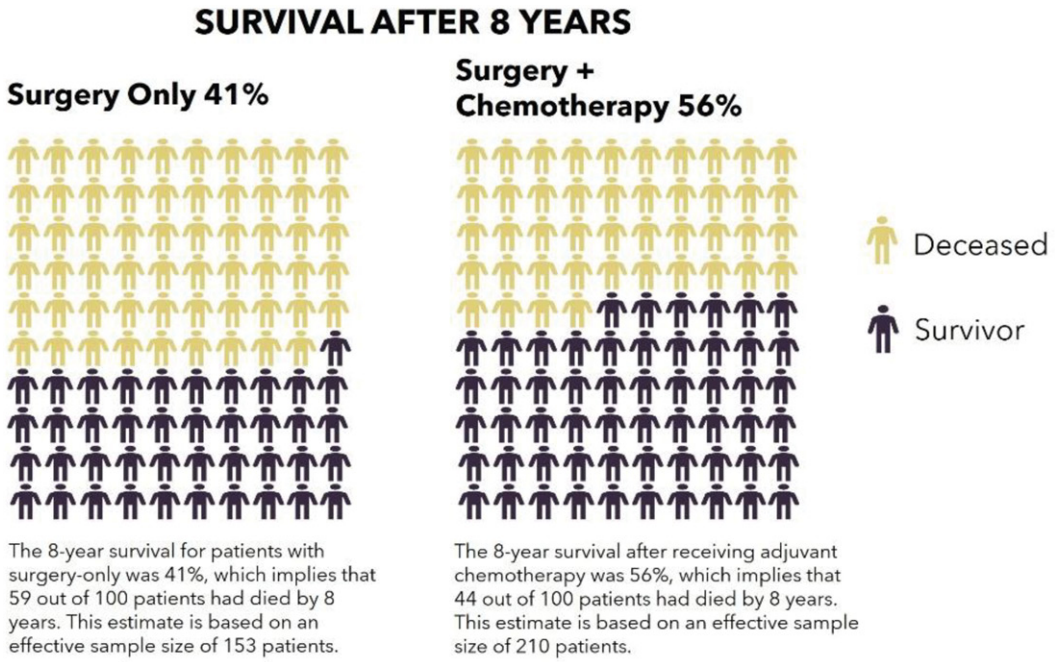

Figure 5. Example of using effective sample size in risk communication for the Colon trial. The 8-year survival is estimated as 41% with surgery only and as 56% with the addition of lev + 5FU chemotherapy after surgery. The effective sample size sizes are 153 and 210 patients, respectively.

patients may prefer simpler expressions of uncertainty such as verbal descriptions rather than numbers (Vromans et al. 2021).

4.3. Effective Sample Size to Guide Study Reporting

Researchers involved in prospective studies with a survival outcome may struggle with the timing of reporting of results. In oncological and other trials, it is common to require a minimum follow-up for the last patient. This causes a gap between inclusion of the last patient and knowledge on the results of a trial, while the results are important to guide further treatment policies. Reporting as early as possible may be pursued, therefore. As illustrated, effective sample size can express the relative information available at earlier versus later time points. For absolute values of sample size, based on a formal power calculation, a higher event rate than anticipated can positively impact the effective sample size. This may allow for earlier analysis and reporting after inclusion of the last patient. In contrast, more censoring by drop out than expected may argue for longer follow-up after the last included patient or increased enrollment for sufficient precision.

We recognize that changes in study design can inflate Type I error rates (Proschan 2009). Further research is required to assess any possible impact on Type I error rates from reporting decisions based on effective sample size.

5. Conclusions

Effective sample size may be valuable to indicate the uncertainty of survival estimates from the Kaplan-Meier estimator, which has the same interpretation as sample size in studies without censoring. It can be shown below Kaplan-Meier plots and risk tables in addition to or instead of the number at risk, where it may yield an intuitive estimate of reliability for estimated survival probabilities until the time of interest. Future research should explore the role of effective sample size to communicate uncertainty, to guide reporting decisions, and to quantify uncertainty from regression or machine learning models with survival outcomes.

Supplementary Materials

The supplementary materials provide a more detailed description of modified effective sample size, including the definition in mathematical formulas. Further, it provides a more extensive description on the simulations used to generate Figure 4, such that it should be reproducible. Finally, it contains some figures of alternative scenarios that were not included in the final manuscript, including an example where both the survival probability and number at risk are identical, but the effective sample size is different and a version of Figure 4 using effective sample size instead of modified effective sample size.

Disclosure Statement

The authors declare no conflicts of interest relevant to this study.

Acknowledgments

4D PICTURE Collaborator authorship: The authors thank the scientific members of the 4D PICTURE consortium who are not authors on this manuscript. Names and affiliations are listed in Supplement 1.

Funding

Funded by the European Union under Horizon Europe Work Programme 101057332. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Health and Digital Executive Agency (HaDEA). Neither

the European Union nor the granting authority can be held responsible for them.

The UK team are funded under the Innovate UK Horizon Europe Guarantee Programme, UKRI Reference Number: 10041120.

Data Availability Statement

The data used in this article is freely available in the survival package in R and at github: *https://github.com/toby-hackmann/ess-kaplan-meier*

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